Arunkumar Rathinam

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Research Interests

Computer Vision for Space Robotics, 6DoF Pose Estimation, Unsupervised Domain Adaptation, Space Situational Awareness (SSA), On-board AI/Edge Computing, Small Body Exploration, and Autonomous Relative Navigation.

Education

Ph.D. in Space Robotics — Oct. 2015 – May. 2019
UNSW Sydney — *Sydney, Australia*
- **Thesis:** Small Body Gravimetry using SLAM-based Autonomous Navigation.

M.Sc. in Space Science and Technology — Oct. 2012 – May. 2015
University of Würzburg — *Würzburg, Germany*
- **Thesis:** Design and development of UWE-4 CubeSat: Integration of electric propulsion, structural analysis, and orbital heating analysis.

B.E. in Mechanical Engineering — Aug. 2003 – May. 2007
Anna University — *Chennai, India*
- First Class with Distinction.

Experience

Research Experience

Research Scientist — Sep. 2024 – Sep.2026
SnT, University of Luxembourg — *Luxembourg*

Research Associate — Sep. 2021 – Aug. 2024
SnT, University of Luxembourg — *Luxembourg*

Research Associate — May 2019 – Sep. 2021
Surrey Space Center, University of Surrey — *United Kingdom*

Industry Experience

Product Engineer — Jan. 2011 – Sep. 2012
SKF Technologies Limited — *India*

Design Engineer — Dec. 2007 – Jan. 2011
TATA Consultancy Services (TCS) — *India*

Research and Project Experience

Project Roles at SnT

DIOSSA Project [ESA GSTP] Phase-1 (24M) & Phase-2 (18M) — 2022 – 2026
Program Lead
- **Leading a team of Researchers** in developing **Deep Learning (DL)** solutions for spacecraft pose estimation.
- Managing full project lifecycle, including **Work Package execution**, monitoring, and progress reporting.
- Successfully delivered all **Phase 1 milestones** and defined the scientific roadmap for Phase 2.
- **Outcomes:** 3 successful **ESA reviews** and 5 conference publications.

Partnership Project with LMO — 2023 – 2026
Technical Lead
- Guiding research strategy on **Unsupervised Domain Adaptation (UDA)** for spacecraft pose estimation, aligning academic research with industrial partner requirements.
- **Co-supervising PhD research** focused on UDA on 6D pose estimation.

MEET-A [FNR-Bridges], ENERGETIC [EU HORIZON] — 2022 – 2025
Researcher
- Provided simulation expertise across multiple research use cases, battery thermal simulation [Energetic], Spacecraft data simulation [MEET-A].

Previous Projects at Surrey

FAIR-SPACE 2019 – 2021
Researcher *University of Surrey, UK*

- Research: Developed DL algorithms for 6DoF spacecraft pose estimation in non-cooperative target scenarios.
- EdgeAI: DL algorithms were evaluated on edge hardware with Hardware-in-the-loop experiments.
- Tech Transfer: Developed ORVIS, an orbital simulation tool to generate training data for relative navigation.
- Output: 2 Journal publications; 1 Book chapter; 2 Conference publications; Supported 1 Start-Up initiative

Space Missions Experience

SnT-AI4SPACE Mission (Deployed) 2022 – 2023
CV and ML Lead *University of Luxembourg, Luxembourg*

- Developed and deployed an onboard AI anomaly detection system, validated on flight hardware.

AUDACITY Mission (Deployed) 2023
Simulation Lead *University of Luxembourg, Luxembourg*

- Designed and deployed an onboard AI algorithm for SSA spacecraft detection SSA in collaboration with LMO.

UWE-4 CubeSat Mission (Deployed) 2015
Student Researcher (Master's Thesis) *University of Würzburg, Germany*

- Developed mechanical design of CubeSat for electric propulsion integration, verified via structural and orbital thermal analysis.

Laboratory Experience

Zero-G Lab 2022 - 2026
Operations Lead (for CVI^2) *University of Luxembourg, Luxembourg*

- Manage end-to-end data-collection campaigns, including req. gathering, test-plan development and execution.
- Coordinate project collaborations with other research groups to streamline workflows and align objectives.
- Coordinate research strategy with equipment and procurement needs, ensuring timely acquisition of resources.
- Outcomes: 3 major 6D Pose datasets, 4 partner data-acquisition campaigns, and 2 journal articles.

Computer Vision Lab 2022 - 2026
Coordinator *University of Luxembourg, Luxembourg*

- Overseeing lab setup, project demos, and equipment procurement.
- Setting up and maintaining CVI^2 GPU Servers within the research group

Orbital Robotics Testbed @ STARLAB Jan. 2020 - May. 2021
Researcher and Coordinator *University of Surrey, UK*

- Coordinated the testbed development for the experimental validation of relative navigation algorithms. link

Floatsat Testbed Jan. 2020 - May. 2021
Research Assistant *University of Würzburg, Germany*

- Designed and implemented a testbed on spherical air bearings for experimental validation of CubeSat ADCS.

Proposals, Research Guidance & Teaching

Grant Proposals & Funding

- **Co-Author:** FNR Bridges U-ADAPT (2025) – **[Funded]**
- **Co-Author:** ESA Proposal DIOSSA Phase-2 (2024) – **[Funded]**
- **Lead Author:** FNR-Junior CORE Sentient-3D (2022) – [Not Funded]
- **Contributor:** FNR CORE DA-GAP (2023); EDF REAL-AI (2022)

Research Guidance & Supervision

- **PhD Co-Supervision:** Nidhal E. Chenni (2024–Present)
  Topic: Domain Adaptation for 6DoF Pose Estimation. Output: 3x Conf. papers (A/A).*
- **PhD Co-Supervision:** Ala Souissi (2025–Present)
  Topic: Leveraging Event Sensing for 6DoF Pose Estimation and Tracking.
- **Master Thesis:** Haytam Qadadri (2023)
  Topic: Spacecraft Pose Estimation using Neuromorphic Sensors. Output: 1x Conf. (A) Publication.*

Teaching Experience

Computer Vision and Image Analysis (CVIA) 2022 – 2026
Course Project & Lab Coordinator *Uni. of Luxembourg*

- Coordinating course projects for MICS & MSTB Master's programs and MSTB lab sessions.

ELEC3106: Engineering Course 2018 – 2019
Teaching Assistant *UNSW Sydney, Australia*

- Facilitated laboratory sessions and exam evaluations for undergraduate engineering students.

Professional Activities

Scientific Event Organization & Service

- **Co-Organizer:** AI4SPACE Workshop @ **CVPR** (2026, 2024), **ECCV** (2022).
- **Lead Organizer:** SPARK Challenge @ **CVPR** (2026, 2024), **ECCV** (2022).
- **Session Chair:** EUVIP 2026.
- **External Advisor:** Neuromorphic Hackathon (2024) @ FORTISS Munich.
- **Program Committee:** IGNSS Conference (2016, 2017).

Peer Review Activities

- **Conferences:** NeurIPS ('26, '25), CVPR ('26, '24), ECCV ('26, '24, '22), ICRA ('26, '25), WACV('25), IROS ('23), AI4SPACE Workshop('26, '24, '22), NEVI workshop ('24).
- **Journals:** Int. Journal of Robotics Research (**IJRR**), IEEE Robotics and Automation Letters (**RA-L**), IEEE Signal Processing Letters (**SPL**), IEEE Trans. on Aerospace and Electronic Systems, Advances in Space Research (**ASR**).

Speaker @ Invited Talks

- **Oct 2023:** ESA CleanSpace Industry Days, ESTEC, Netherlands.
  "DL-based spacecraft pose estimation: Opportunities and challenges for future in-orbit servicing missions."
- **Sep 2020:** Fairspace Workshop, Liverpool, UK.
  "On-orbit relative pose estimation of a non-cooperative target using convolutional neural networks."
- **Oct 2018:** 17th Space Generation Congress, Bremen, Germany.
  "Reusable CubeSat constellation for asteroid rover navigation and communication."
- **Sep 2017:** 3rd Off Earth Mining Forum, UNSW Sydney, Australia.
  "Autonomous spacecraft navigation near an asteroid."

Technical Skills

- **Deep Learning:** PyTorch, PyTorch Lightning, TensorFlow, Hydra, OpenCV, NumPy.
- **Robotics & Simulation:** ROS, PANGU (ESA), Unreal Engine 4, Blender.
- **Programming:** Python, C++, MATLAB, Julia.
- **Embedded AI:** NVIDIA Jetson (Nano / Xavier NX), Arduino, Edge Inference Optimization.
- **Mechanical Engineering: CAD:** Creo, CATIA V5, SolidWorks, Inventor; **FEM/FEA:** NASTRAN, COMSOL.
- **DevOps & Tools:** Git, Docker, Windchill PDM, Nginx, Gunicorn.

Honors & Awards

- **Winner, Move an Asteroid Technical Paper Competition (2018)**
  Space Generation Advisory Council (SGAC), Bremen.
- **IAC Travel Grant / Sponsorship (2017)**
  Awarded by the National Space Society of Australia to attend the International Astronautical Congress.
- **UNSW Departmental PhD Scholarship (2015)**
  Full tuition coverage plus living stipend for duration of candidature.
- **Erasmus Mobility Grant (2013)**
  Competitive grant for Master's exchange year.
- **Top Achiever Award — SKF (2012)**
  Ranked 1st in performance assessment during tenure at SKF India.

Research Publications

- Full list available in the **Google Scholar** → `link`

Journal Articles

1. **A. Rathinam** et al., "Hybrid Attention for Robust RGB-T Pedestrian Detection in Real-World Conditions," *IEEE Robotics and Automation Letters*, vol. 10, no. 1, Jan. 2025. DOI: 10.1109/LRA.2024.3504296.
2. A. E. R. Shabayek, T. Amietszajew, **A. Rathinam**, and D. Aouada, "Thermal Anomaly Detection for Battery Safety in Electric Vehicles," 2025. DOI: 10.2139/ssrn.5579224.
3. A. Shabayek, **A. Rathinam**, M. Ruthven, D. Aouada, and T. Amietszajew, "AI-enabled thermal monitoring of commercial (PHEV) Li-ion pouch cells with Feature-Adapted Unsupervised Anomaly Detection," *Journal of Power Sources*, vol. 629, Feb. 2025. DOI: 10.1016/j.jpowsour.2024.235982.
4. J. Thoemel et al., "Lean Demonstration of On-Board Thermal Anomaly Detection Using Machine Learning," *Aerospace*, vol. 11, no. 7, Jun. 2024. DOI: 10.3390/aerospace11070523.
5. M. Olivares-Mendez et al., "Establishing a Multi-Functional Space Operations Emulation Facility: Insights from the Zero-G Lab," 2023. DOI: 10.2139/ssrn.4602588.
6. M. Olivares-Mendez et al., "Zero-G Lab: A multi-purpose facility for emulating space operations," *Journal of Space Safety Engineering*, vol. 10, no. 4, Dec. 2023. DOI: 10.1016/j.jsse.2023.09.003.
7. L. Pauly, W. Rharbaoui, C. Shneider, **A. Rathinam**, V. Gaudillière, and D. Aouada, "A survey on deep learning-based monocular spacecraft pose estimation: Current state, limitations and prospects," *Acta Astronautica*, vol. 212, Nov. 2023. DOI: 10.1016/j.actaastro.2023.08.001.
8. L. Pauly et al., "Lessons from a space lab – an image acquisition perspective," *International Journal of Aerospace Engineering*, vol. 2023, 2023. DOI: 10.1155/2023/9944614.
9. R. B. Ashith Shyam et al., "Autonomous robots for space: Trajectory learning and adaptation using imitation," *Frontiers in Robotics and AI*, vol. 8 - 2021, 2021. DOI: 10.3389/frobt.2021.638849.
10. Z. Hao, R. B. A. Shyam, **A. Rathinam**, and Y. Gao, "Intelligent Spacecraft Visual GNC Architecture With the State-Of-the-Art AI Components for On-Orbit Manipulation," *Frontiers in Robotics and AI*, vol. 8 - 2021, 2021. DOI: 10.3389/frobt.2021.639327.
11. **A. Rathinam** and A. G. Dempster, "Octree-Based Mascon Model for Small Body Gravity Fields," *Journal of Guidance, Control, and Dynamics*, vol. 42, no. 11, Nov. 2019. DOI: 10.2514/1.G004008.

Conference Proceedings

1. N. Eddine Chenni, **A. Rathinam**, and D. Aouada, "Eclipse: A spacecraft rendezvous trajectories dataset with controlled in-orbit lighting conditions," in *Proceedings of the Fortieth Annual Conference on Neural Information Processing Systems 2026*, (accepted), Nov. 2027.
2. N. Eddine Chenni, **A. Rathinam**, and D. Aouada, "Scope: Saliency-guided cross-domain alignment for 6d object pose estimation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision 2027*, (accepted), IEEE, Jan. 2027.
3. N. Eddine Chenni, **A. Rathinam**, and D. Aouada, "Unsupervised domain adaptation for sim-to-real object pose estimation with contrastive alignment and pseudo-label refinement," in *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, (accepted), IEEE, Oct. 2026.
4. S. Hicsonmez et al., "Annotation free spacecraft detection and segmentation using vision language models," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria: IEEE, Jun. 2026.

5. **A. Rathinam**, J. Lecomte, J. Reelsen, G. Lenz, A. von Arnim, and D. Aouada, "Efficient onboard spacecraft pose estimation with event cameras and neuromorphic hardware," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026 workshops*, IEEE, Jun. 2026.

6. S. Hicsonmez, A. E. R. Shabayek, **A. Rathinam**, and D. Aouada, "Domain adaptive object detection for space applications with real-time constraints," in *18th Symposium on Advanced Space Technologies in Robotics and Automation*, Leiden, Netherlands: ESA, Oct. 2025.

7. I. P. Singh, N. E. Chenni, A. E. R. Shabayek, **A. Rathinam**, and D. Aouada, "Bridging the synthetic-real gap: Supervised domain adaptation for robust spacecraft 6-dof pose estimation," in *18th Symposium on Advanced Space Technologies in Robotics and Automation*, Leiden, Netherlands: ESA, Oct. 2025.

8. J. A. Sosa Martinez, D. Pineau, **A. Rathinam**, A. E. R. Shabayek, and D. Aouada, "Motion aware vit-based framework for monocular 6-dof spacecraft pose estimation," in *18th Symposium on Advanced Space Technologies in Robotics and Automation*, Leiden, Netherlands: ESA, Oct. 2025.

9. **A. Rathinam**, H. Qadadri, and D. Aouada, "SPADES: A Realistic Spacecraft Pose Estimation Dataset using Event Sensing," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan: IEEE, May 2024. DOI: 10.1109/ICRA57147.2024.10611231.

10. I. P. Singh, E. Ghorbel, A. Kacem, **A. Rathinam**, and D. Aouada, "Discriminator-free unsupervised domain adaptation for multi-label image classification," in *2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024. DOI: 10.1109/WACV57701.2024.00389.

11. V. Gaudillière, L. Pauly, **A. Rathinam**, A. G. Sanchez, M. A. Musallam, and D. Aouada, "3D-Aware Object Localization using Gaussian Implicit Occupancy Function," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, MI, USA: IEEE, Oct. 2023. DOI: 10.1109/IROS55552.2023.10342399.

12. M. A. Musallam, **A. Rathinam**, V. Gaudillière, M. O. d. Castillo, and D. Aouada, "CubeSat-CDT: A Cross-Domain Dataset for 6-DoF Trajectory Estimation of a Symmetric Spacecraft," in *Proceedings of the 17th European Conference on Computer Vision Workshops (ECCVW 2022)*, Springer Nature Switzerland, 2023, pp. 112–126. DOI: 10.1007/978-3-031-25056-9_8.

13. **A. Rathinam**, V. Gaudillière, L. Pauly, and D. Aouada, "Pose estimation of a known texture-less space target using convolutional neural networks," in *73rd International Astronautical Congress*, Paris, France, Sep. 2022.

14. **A. Rathinam** and Y. Gao, "On-orbit relative navigation near a known target using monocular vision and convolutional neural networks for pose estimation," in *International Symposium on Artificial Intelligence, Robotics and Automation in Space (iSAIRAS)*, Virtual, 2020.

15. A. Shyam, **A. Rathinam**, and Z. Hao, "Trajectory tracking and control of multiple robot arms on a free-floating spacecraft for debris removal," in *3rd UK-RAS Conference*, Leicestershire, UK, 2020.

16. **A. Rathinam**, "Orbital analysis of a reusable cubesat platform to relay communications for rovers on an asteroid," in *Move an Asteroid Competition – Space Generation Congress*, Bremen, Germany, 2018.

17. **A. Rathinam** and A. G. Dempster, "Vision based state estimation using a graph-SLAM approach for proximity operations near an asteroid," in *69th International Astronautical Congress*, Bremen, Germany, 2018.

18. **A. Rathinam** and A. G. Dempster, "3D reconstruction of an asteroid shape using visual SLAM for autonomous navigation," in *17th Australian Space Research Conference (ASRC)*, Sydney, Australia, 2017.

19. **A. Rathinam** and A. G. Dempster, "Monocular vision based simultaneous localization and mapping for close proximity navigation near an asteroid," in *68th International Astronautical Congress*, Adelaide, Australia, 2017.

20 **A. Rathinam** and A. G. Dempster, "Effective Utilization of Space Service Volume through Combined GNSS," in *Proceedings of the 29th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2016)*, Portland, Oregon, Nov. 2016. DOI: 10.33012/2016.14747.

21 **A. Rathinam** and A. G. Dempster, "Geometrical assessment of Multi-GNSS prospects for Space Service Volume," in *16th Australian Space Research Conference (ASRC)*, Melbourne, Australia, 2016, pp. 199–206.

22 **A. Rathinam** and A. G. Dempster, "Multi-GNSS for Space Service Volume," in *International Global Navigation Satellite Systems (IGNSS) 2016*, Sydney, Australia, 2016.

23 **A. Rathinam**, P. Bangert, D. Bock, M. Pietzka, M. Tajmar, and K. Schilling, "Structural Design, Integration of Micro Propulsion units and Thermal Analysis of University of Wuerzburg's Experiment (UWE) Satellite Platform," in *6th European Conference for AeroSpace Sciences (EUCASS)*, 2015.

Books and Chapters

1 **A. Rathinam**, Z. Hao, and Y. Gao, "Autonomous visual navigation for spacecraft on-orbit operations," in *Space Robotics and Autonomous Systems: Technologies, advances and applications*, ch. Chapter 5, pp. 125–157. DOI: 10.1049/PBCE131E_ch5.

Datasets

1 **A. Rathinam**, M. A. Mohamed Ali, V. Gaudilliere, and D. Aouada, *SPARK 2024: Datasets for Spacecraft Semantic Segmentation and Spacecraft Trajectory Estimation*, Feb. 2024. DOI: 10.5281/ZENODO.10908214.

2 **A. Rathinam**, V. Gaudilliere, M. A. Mohamed Ali, M. Ortiz Del Castillo, L. Pauly, and D. Aouada, *SPARK 2022 Dataset : Spacecraft Detection and Trajectory Estimation*, Jun. 2022. DOI: 10.5281/ZENODO.6599762.